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Online-Appendix

Appendix A. An example of virtual avatar streaming

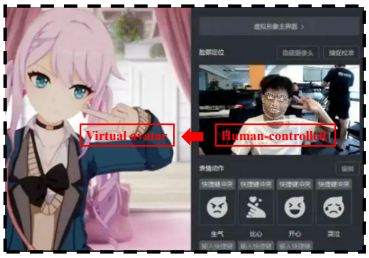


Fig. A1. An example of virtual avatar streaming.

Appendix B. Summary of notations

Table A1 summarizes key notation in the model.

Appendix C. Equilibrium results

Lemma A1. (equilibrium results under human streaming model). *Under the human streaming model, the platform's optimal creative reward is* $\alpha^{H*} = \frac{\gamma-(1-2\xi)\beta-k\underline{q}^H}{2}$, *with an optimal profit* $R^{H*} = \frac{(\beta+\gamma+k\underline{q}^H)^2}{4k}$. *The streamer's optimal creative effort is* $e^{H*} = \frac{\beta+\gamma-k\underline{q}^H}{2k}$, *with an optimal profit* $S^{H*} = \frac{(\beta+\gamma-k\underline{q}^H)(\beta+\gamma+3k\underline{q}^H)}{8k}$. *Viewers' optimal content consumption is* $\omega^{H*} = \frac{\beta+\gamma+k\underline{q}^H}{2k}$, *with a perceived ELS content quality* $q^{H*} = \frac{\beta+\gamma+k\underline{q}^H}{2k}$.

Proof.

Using backward induction, we start the analysis from Stage 3. Taking the first-order condition (F.O.C) of Q^H with respect to ω^H, we have $\frac{\partial Q^H}{\partial \omega^H} = q^H - \omega^H$. Setting $\frac{\partial Q^H}{\partial \omega^H} = 0$, we have $\omega^H = q^H$. Moreover, $\frac{\partial^2 Q^H}{\partial \omega^{H2}} = -1 < 0$ ensures that ω^H maximizes Q^H. Next, we substitute $\omega^H = q^H$ into the streamer's problem in Stage 2. Taking the F.O.C of S^H with respect to e^H, we have $\frac{\partial S^H}{\partial e^H} = \alpha^H + (1-\xi)\beta - ke^H$. Setting

Table A1. Summary of notations

Notation	Definition
Parameters	
γ	Marginal traffic revenue of the platform, $\gamma > \alpha$
ξ	The platform's commission rate on viewer tips, $\xi \in (0,1)$
β	The streamer's potential tipping revenue, $\beta > 0$
k	Cost coefficient of content creation, $k > 1$
$\underline{q}$	Baseline quality of the ELS, $\underline{q} > 0$
c	Fixed cost of adopting a virtual avatar, $c > 0$
λ	Adverse effect of virtual avatar streaming, $\lambda \in (0,1)$
Intermediate variables	
q	Viewers' perceived ELS content quality
Q	Viewers' utility function
R	The platform's profit function
S	The streamer's profit function
Decision variables	
ω	Viewers' content consumption
α	The platform's creative rewards
e	The streamer's creative effort
Superscripts	
H	Human streaming
V	virtual avatar streaming

$\frac{\partial S^H}{\partial e^H} = 0$, we have $e^H = \frac{\alpha^H + (1-\xi)\beta}{k}$. In addition, $\frac{\partial^2 S^H}{\partial e^{H2}} = -k < 0$ ensures that e^H maximizes S^H. Finally, we consider the platform's problem in Stage 1. Taking the F.O.C of R^H with respect to α^H, we have $\frac{\partial R^H}{\partial \alpha^H} = \frac{\gamma + (2\xi - 1)\beta - k\underline{q}^H - 2\alpha^H}{k}$. Setting $\frac{\partial R^H}{\partial \alpha^H} = 0$, we have $\alpha^{H*} = \frac{\gamma - (1-2\xi)\beta - k\underline{q}^H}{2}$. In addition, $\frac{\partial^2 R^H}{\partial \alpha^{H2}} = -\frac{2}{k} < 0$ ensures α^{H*} maximizes R^H. Furthermore, we obtain $e^{H*} = \frac{\beta + \gamma - k\underline{q}^H}{2k}$ and $\omega^{H*} = \frac{\beta + \gamma + k\underline{q}^H}{2k}$. By substituting the equilibrium results into the platform's and the streamer's utility functions, Lemma A1 is proved.

It is worth noting that the following sufficient conditions are imposed to ensure that $0 < \alpha^{H*} < \gamma$. (1) $\xi < \frac{\beta + k\underline{q}^H}{2\beta}$, meaning that the commission rate charged by the platform on the streamer's tipping revenue is below a certain threshold. (2) $\gamma > \beta(1-2\xi) + k\underline{q}^H$, meaning that the marginal traffic revenue generated for the platform is sufficiently large. These technical assumptions guarantee that the platform has an incentive to provide creative rewards to the streamer. □

Lemma A2. (equilibrium results under the virtual avatar streaming model). *Under virtual avatar streaming model, the platform's optimal creative reward is* $\alpha^{V*} = \frac{\lambda^2[\gamma - (1-2\xi)\beta] - \underline{q}^V}{2\lambda^2}$, *with an optimal profit* $R^{V*} = \frac{\left[\lambda^2(\beta+\gamma) + \underline{q}^V\right]^2}{4\lambda^2}$. *The*

streamer's optimal creative effort is $e^{V*} = \frac{\lambda^2(\beta+\gamma)-\underline{q}^V}{2\lambda}$*, with an optimal profit* $S^{V*} = \frac{\lambda^4(\beta+\gamma)^2-2\lambda^2\underline{q}^V(4-\beta-\gamma)-3(\underline{q}^V)^2}{8\lambda^2}$*. Viewers' optimal content consumption is* $\omega^{V*} = \frac{\underline{q}^V+\lambda^2(\beta+\gamma)}{2}$*, with a perceived ELS content quality* $q^{V*} = \frac{\lambda^2(\beta+\gamma)+\underline{q}^V}{2}$.

Proof. Similar to the proof of Lemma A1. Taking the F.O.C. of Q^V with respect to ω^V, we have $\frac{\partial Q^V}{\partial \omega^V} = q^V - \omega^V$. Setting $\frac{\partial Q^V}{\partial \omega^V} = 0$, we obtain $\omega^V = q^V$. Moreover, $\frac{\partial^2 Q^V}{\partial (\omega^V)^2} = -1 < 0$ ensures that ω^V maximizes Q^V. Next, we substitute $\omega^V = q^V = \underline{q}^V + \lambda e^V$ into the streamer's problem in Stage 2. Taking the F.O.C. of S^V with respect to e^V, we have $\frac{\partial S^V}{\partial e^V} = \lambda[\alpha^V + (1-\xi)\beta] - e^V$. Setting $\frac{\partial S^V}{\partial e^V} = 0$, we obtain $e^V = \lambda[\alpha^V + (1-\xi)\beta]$. In addition, $\frac{\partial^2 S^V}{\partial (e^V)^2} = -1 < 0$ ensures that e^V maximizes S^V. Finally, we consider the platform's problem in Stage 1. Substituting the streamer's optimal response into viewers' content consumption yields $\omega^V = \underline{q}^V + \lambda^2[\alpha^V + (1-\xi)\beta]$. Taking the F.O.C. of R^V with respect to α^V, we have $\frac{\partial R^V}{\partial \alpha^V} = \lambda^2[\gamma + (2\xi-1)\beta - 2\alpha^V] - \underline{q}^V$. Setting $\frac{\partial R^V}{\partial \alpha^V} = 0$, we obtain $\alpha^{V*} = \frac{\lambda^2[\gamma-(1-2\xi)\beta]-\underline{q}^V}{2\lambda^2}$. In addition, $\frac{\partial^2 R^V}{\partial (\alpha^V)^2} = -2\lambda^2 < 0$ ensures that α^{V*} maximizes R^V. Furthermore, we obtain $e^{V*} = \frac{\lambda^2(\beta+\gamma)-\underline{q}^V}{2\lambda}$, and $\omega^{V*} = q^{V*} = \frac{\underline{q}^V+\lambda^2(\beta+\gamma)}{2}$. By substituting these equilibrium decisions into the platform's and the streamer's objective functions and using $\underline{q}^V = c$, we obtain $R^{V*} = \frac{\left[\lambda^2(\beta+\gamma)+\underline{q}^V\right]^2}{4\lambda^2}$, and $S^{V*} = \frac{\lambda^4(\beta+\gamma)^2-2\lambda^2\underline{q}^V(4-\beta-\gamma)-3(\underline{q}^V)^2}{8\lambda^2}$. Lemma A2 is proved.

It is worth noting that the following sufficient conditions are imposed to ensure that $0 < \alpha^{V*} < \gamma$. (1)$\xi < \frac{\beta\lambda^2+\underline{q}^V}{2\beta\lambda^2}$, and (2)$\gamma > \beta(1-2\xi) + \frac{\underline{q}^V}{\lambda^2}$. □

Appendix D. Proofs

D.1. *Proof of Proposition 1*

Proof. From Lemmas A1 and A2, we have $\alpha^{V*} - \alpha^{H*} = \frac{k\lambda^2\underline{q}^H-\underline{q}^V}{2\lambda^2}$. Define $\underline{q}_1^V = k\lambda^2\underline{q}^H$. Since $2\lambda^2 > 0$, we have $\alpha^{V*} > \alpha^{H*}$ when $\underline{q}^V < \underline{q}_1^V$, whereas $\alpha^{V*} < \alpha^{H*}$ when $\underline{q}^V > \underline{q}_1^V$. Under the virtual avatar streaming model, the interior condition $\alpha^{V*} > 0$ requires $\underline{q}^V < \lambda^2[\gamma - (1-2\xi)\beta]$. The threshold $\underline{q}_1^V$ lies within this feasible region if $\underline{q}_1^V < \lambda^2[\gamma-(1-2\xi)\beta]$, which is equivalent to $\underline{q}^H < \frac{\gamma-(1-2\xi)\beta}{k}$. This condition is exactly the interior condition $\alpha^{H*} > 0$ under the human streaming model. Therefore, the maintained interior conditions automatically ensure that $\underline{q}_1^V$ lies within the common feasible region, and no additional threshold for $\underline{q}^H$ is required. □

D.2. *Proof of Proposition 2*

Proof. We have $q^{V*} - q^{H*} = \frac{1}{2}[\underline{q}^V - \underline{q}^H - (\beta+\gamma)(\frac{1}{k} - \lambda^2)]$. Define $\underline{q}_2^V = \underline{q}^H + (\beta+\gamma)(\frac{1}{k} - \lambda^2)$. Thus, $q^{V*} > q^{H*}$ when $\underline{q}^V > \underline{q}_2^V$, whereas $q^{V*} < q^{H*}$ when $\underline{q}^V < \underline{q}_2^V$.

Under the virtual avatar streaming model, the interior condition $\alpha^{V*} > 0$ requires $\underline{q}^V < \lambda^2[\gamma - (1-2\xi)\beta]$. Hence, the condition $\underline{q}^V > \underline{q}_2^V$ is feasible only if $\underline{q}_2^V < \lambda^2[\gamma - (1-2\xi)\beta]$, which is equivalent to $\underline{q}^H < \underline{q}_1^H$, where $\underline{q}_1^H = 2\lambda^2(\gamma + \xi\beta) - \frac{\beta+\gamma}{k}$. Therefore, when $\underline{q}^H < \underline{q}_1^H$, the comparison depends on whether $\underline{q}^V$ is above or below $\underline{q}_2^V$. When $\underline{q}^H > \underline{q}_1^H$, $\underline{q}_2^V$ lies outside the feasible region, implying $q^{V*} < q^{H*}$ for all feasible $\underline{q}^V$. □

D.3. *Proof of Proposition 3*

Proof. We have $R^{V*} - R^{H*} = \frac{1}{4}[\frac{\lambda^2(\beta+\gamma)+\underline{q}^V}{\lambda} - \frac{\beta+\gamma+k\underline{q}^H}{\sqrt{k}}][\frac{\lambda^2(\beta+\gamma)+\underline{q}^V}{\lambda} + \frac{\beta+\gamma+k\underline{q}^H}{\sqrt{k}}]$. Because the second term is positive, the sign of $R^{V*} - R^{H*}$ is determined by the first term. Define $\underline{q}_3^V = \lambda\sqrt{k}\underline{q}^H + (\beta+\gamma)(\frac{\lambda}{\sqrt{k}} - \lambda^2)$. Thus, $R^{V*} > R^{H*}$ when $\underline{q}^V > \underline{q}_3^V$, whereas $R^{V*} < R^{H*}$ when $\underline{q}^V < \underline{q}_3^V$. Under the virtual avatar streaming model, the interior condition $\alpha^{V*} > 0$ requires $\underline{q}^V < \lambda^2[\gamma - (1-2\xi)\beta]$. Hence, the condition $\underline{q}^V > \underline{q}_3^V$ is feasible only if $\underline{q}_3^V < \lambda^2[\gamma - (1-2\xi)\beta]$, which is equivalent to $\underline{q}^H < \underline{q}_2^H$, where $\underline{q}_2^H = \frac{2\lambda(\gamma+\xi\beta)}{\sqrt{k}} - \frac{\beta+\gamma}{k}$. Therefore, when $\underline{q}^H < \underline{q}_2^H$, the comparison depends on whether $\underline{q}^V$ is above or below $\underline{q}_3^V$. When $\underline{q}^H > \underline{q}_2^H$, the threshold $\underline{q}_3^V$ lies outside the feasible region, implying $R^{V*} < R^{H*}$ for all feasible $\underline{q}^V$. □

D.4. *Proof of Proposition 4*

Proof. We have $S^{V*} - S^{H*} = \frac{1}{8}[3k(\underline{q}^H)^2 - 2(\beta+\gamma)\underline{q}^H + (\beta+\gamma)^2(\lambda^2 - \frac{1}{k}) - 2(4-\beta-\gamma)\underline{q}^V - \frac{3(\underline{q}^V)^2}{\lambda^2}]$. It is easy to find that $\frac{\partial(S^{V*}-S^{H*})}{\partial \underline{q}^V} = -\frac{4-\beta-\gamma}{4} - \frac{3\underline{q}^V}{4\lambda^2} < 0$. Thus, $S^{V*} - S^{H*}$ decreases in $\underline{q}^V$. At $\underline{q}^V = 0$, the equation $S^{V*} - S^{H*} = 0$ becomes $3k(\underline{q}^H)^2 - 2(\beta+\gamma)\underline{q}^H + (\beta+\gamma)^2(\lambda^2 - \frac{1}{k}) = 0$. Define the positive root as $\underline{q}_3^H = \frac{(\beta+\gamma)[1+\sqrt{4-3k\lambda^2}]}{3k}$. Therefore, when $\underline{q}^H < \underline{q}_3^H$, we have $S^{V*} - S^{H*} < 0$ for all positive $\underline{q}^V$. When $\underline{q}^H > \underline{q}_3^H$, we have $S^{V*} - S^{H*} > 0$ at $\underline{q}^V = 0$. Because $S^{V*} - S^{H*}$ decreases in $\underline{q}^V$, there exists a unique positive threshold satisfying $S^{V*} = S^{H*}$. Solving $S^{V*} - S^{H*} = 0$ with respect to $\underline{q}^V$ gives $\underline{q}_4^V = \frac{\lambda^2}{3}\{\beta + \gamma - 4 + \sqrt{(4-\beta-\gamma)^2 + \frac{3}{\lambda^2}[3k(\underline{q}^H)^2 - 2(\beta+\gamma)\underline{q}^H + (\beta+\gamma)^2(\lambda^2 - \frac{1}{k})]}\}$. Hence, when $\underline{q}^H > \underline{q}_3^H$, we have $S^{V*} > S^{H*}$ if $\underline{q}^V < \underline{q}_4^V$, whereas $S^{V*} < S^{H*}$ if $\underline{q}^V > \underline{q}_4^V$. □

D.5. *Proof of Proposition 5*

Proof. From Lemma A3, we have $\frac{\partial e_i^{H*}}{\partial \eta^H} = \frac{(\beta+\gamma)\eta^H(m-1)[2+(m-2)\eta^H]}{2k(1-\eta^H)^2[1+\eta^H(m-1)]^2} > 0$, $\frac{\partial \alpha^{H*}}{\partial \eta^H} = \frac{k\underline{q}^H\eta^H(m-1)[2+(m-2)\eta^H]}{2[1+\eta^H(m-2)]^2} > 0$, $\frac{\partial e_j^{V*}}{\partial \eta^V} = \frac{\lambda(\beta+\gamma)\eta^V(n-m-1)[2+(n-m-2)\eta^V]}{2(1-\eta^V)^2[1+\eta^V(n-m-1)]^2} > 0$, and $\frac{\partial \alpha^{V*}}{\partial \eta^V} = \frac{\underline{q}^V\eta^V(n-m-1)[2+(n-m-2)\eta^V]}{2\lambda^2[1+\eta^V(n-m-2)]^2} > 0$. □

D.6. *Proof of Proposition 6*

The results follow from the arguments made in the main text.

Appendix E. Some examples of acquiring virtual avatars

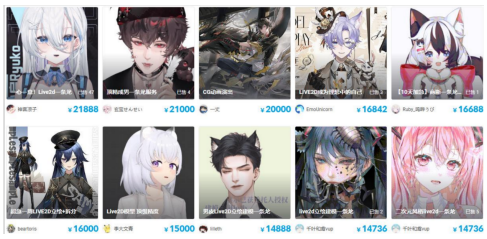


Fig. A2. High-end virtual avatars.

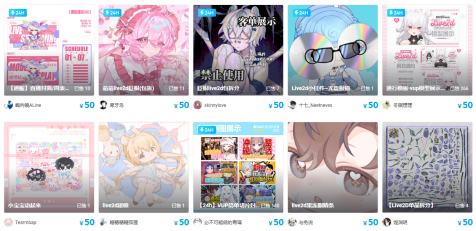


Fig. A3. Low-end virtual avatars.

Appendix F. Analysis of Extensions

F.1. *Multiple streamers in a competitive setting*

F.1.1. *Equilibrium results*

Lemma A3. *The equilibrium results are as follow:*

$$e_i^{H*} = \frac{(\beta+\gamma)[1+\eta^H(m-2)] - k\underline{q}^H(1-\eta^H)[1+\eta^H(m-1)]}{2k(1-\eta^H)[1+\eta^H(m-1)]}$$

$$e_j^{V*} = \frac{\lambda^2(\beta+\gamma)[1+\eta^V(n-m-2)] - \underline{q}^V(1-\eta^V)[1+\eta^V(n-m-1)]}{2\lambda(1-\eta^V)[1+\eta^V(n-m-1)]}$$

$$q_i^{H*} = \frac{k\underline{q}^H(1-\eta^H)[1+\eta^H(m-1)] + (\beta+\gamma)[1+\eta^H(m-2)]}{2k(1-\eta^H)[1+\eta^H(m-1)]}$$

$$q_j^{V*} = \frac{\underline{q}^V(1-\eta^V)[1+\eta^V(n-m-1)] + \lambda^2(\beta+\gamma)[1+\eta^V(n-m-2)]}{2(1-\eta^V)[1+\eta^V(n-m-1)]}$$

$$\alpha^{H*} = \frac{[\gamma-(1-2\xi)\beta][1+\eta^H(m-2)] - k\underline{q}^H(1-\eta^H)[1+\eta^H(m-1)]}{2[1+\eta^H(m-2)]}$$

$$\alpha^{V*} = \frac{\lambda^2[\gamma-(1-2\xi)\beta][1+\eta^V(n-m-2)] - \underline{q}^V(1-\eta^V)[1+\eta^V(n-m-1)]}{2\lambda^2[1+\eta^V(n-m-2)]}$$

$$\begin{aligned} S_i^{H*} =& \frac{\{\beta+\gamma-\frac{k\underline{q}^H(1-\eta^H)[1+\eta^H(m-1)]}{1+\eta^H(m-2)}\}}{4[1+\eta^H(m-1)]} \times \{\underline{q}^H + \frac{(\beta+\gamma)[1+\eta^H(m-2)]}{k(1-\eta^H)[1+\eta^H(m-1)]}\} \\ & - \frac{k}{8}\{\frac{(\beta+\gamma)[1+\eta^H(m-2)]}{k(1-\eta^H)[1+\eta^H(m-1)]} - \underline{q}^H\}^2 \end{aligned}$$

$$\begin{aligned} S_j^{V*} =& \frac{\{\beta+\gamma-\frac{\underline{q}^V(1-\eta^V)[1+\eta^V(n-m-1)]}{\lambda^2[1+\eta^V(n-m-2)]}\}}{4[1+\eta^V(n-m-1)]} \times \{\underline{q}^V + \frac{\lambda^2(\beta+\gamma)[1+\eta^V(n-m-2)]}{(1-\eta^V)[1+\eta^V(n-m-1)]}\} \\ & - \frac{1}{8}\{\frac{\lambda(\beta+\gamma)[1+\eta^V(n-m-2)]}{(1-\eta^V)[1+\eta^V(n-m-1)]} - \frac{\underline{q}^V}{\lambda}\}^2 - \underline{q}^V \end{aligned}$$

$$R^{H*} = \frac{m\left\{(\beta+\gamma)[1+\eta^H(m-2)] + k\underline{q}^H(1-\eta^H)[1+\eta^H(m-1)]\right\}^2}{4k(1-\eta^H)[1+\eta^H(m-1)]^2[1+\eta^H(m-2)]}$$

$$R^{V*} = \frac{(n-m)\left\{\lambda^2(\beta+\gamma)[1+\eta^V(n-m-2)] + \underline{q}^V(1-\eta^V)[1+\eta^V(n-m-1)]\right\}^2}{4\lambda^2(1-\eta^V)[1+\eta^V(n-m-1)]^2[1+\eta^V(n-m-2)]}$$

Proof. Similar to the proof of our main model. Using backward induction, we start the analysis from Stage 3. Taking the F.O.C of Q with respect to ω_i^H and ω_j^V, we have $\frac{\partial Q}{\partial \omega_i^H} = q_i^H - \omega_i^H - \eta^H \sum_{l=1}^{m} \omega_k^H$ and $\frac{\partial Q}{\partial \omega_j^V} = q_j^V - \omega_j^V - \eta^V \sum_{r=m+1}^{n} \omega_r^V$. Where $l \neq i$ and $r \neq j$. Setting $\frac{\partial Q}{\partial \omega_i^H} = 0$ and $\frac{\partial Q}{\partial \omega_j^V} = 0$, we have $\omega_i^H = q_i^H - \eta^H \sum_{l=1}^{m} \omega_l^H$ and $\omega_j^V = q_j^V - \eta^V \sum_{r=m+1}^{n} \omega_r^V$. On this basis, we have $\sum_{k=1}^{m} \omega_l^H = \sum_{l=1}^{m} q_k^H - \eta^H(m-1) \sum_{l=1}^{m} \omega_l^H$ and $\sum_{r=m+1}^{n} \omega_r^V = \sum_{r=m+1}^{n} q_r^V - \eta^V(n-m-1) \sum_{r=m+1}^{n} \omega_r^V$. By solving the above equations, we have $\sum_{l=1}^{m} \omega_l^H = \frac{\sum_{l=1}^{m} q_k^H}{1+\eta^H(m-1)}$. Therefore, $\omega_i^H = q_i^H - \eta^H \sum_{l=1}^{m} \omega_l^H = q_i^H - \eta^H[\frac{\sum_{l=1}^{m} q_l^H}{1+\eta^H(m-1)} - \omega_i^H]$ and $\omega_j^V = q_j^V - \eta^V \sum_{r=m+1}^{n} \omega_r^V = q_j^V - \eta^V[\frac{\sum_{j=m+1}^{n} q_j^V}{1+\eta^V(n-m-1)} -$

$\omega_j^V]$. By solving the above equations, we have $\omega_i^H = \frac{q_i^H[1+\eta^H(m-2)]-\eta^H\sum_{l=1}^{m} q_l^H}{(1-\eta^H)[1+\eta^H(m-1)]}$ and $\omega_j^V = \frac{q_j^V[1+\eta^V(n-m-2)]-\eta^V\sum_{r=m+1}^{n} q_r^V}{(1-\eta^V)[1+\eta^V(n-m-1)]}$. Moreover, $\frac{\partial^2 Q}{\partial \omega_i^{H2}} = -1 < 0$ and $\frac{\partial^2 Q}{\partial \omega_j^{V2}} = -1 < 0$ ensure the optimality of the above results. The remaining proof steps are the same as in the main model and are not repeated here.

It is worth noting that the following sufficient conditions are imposed to ensure that $0 < \alpha^{H*} < \gamma$ and $0 < \alpha^{V*} < \gamma$. (1) $\gamma > (1-2\xi)\beta$, (2) $0 < \underline{q}^H < \frac{[\gamma-(1-2\xi)\beta][1+\eta^H(m-2)]}{k(1-\eta^H)[1+\eta^H(m-1)]}$, and (3) $0 < \underline{q}^V < \frac{\lambda^2[\gamma-(1-2\xi)\beta][1+\eta^V(n-m-2)]}{(1-\eta^V)[1+\eta^V(n-m-1)]}$. □

F.1.2. *Robustness test*

We next examine whether the main findings in Propositions 1–4 remain valid in this setting.[a]

Proposition A1.

(a) (Creative reward). *There exists a threshold $\underline{q}_{1c}^V$ such that: (1) When $\underline{q}^V < \underline{q}_{1c}^V$, the platform tends to offer a higher creative reward to virtual avatar streamers, i.e.,$\alpha^{V*} > \alpha^{H*}$; (2) When $\underline{q}^V > \underline{q}_{1c}^V$, the platform tends to offer a lower creative reward to virtual avatar streamers, i.e., $\alpha^{V*} < \alpha^{H*}$.*

(b) (Viewers' perceived ELS content quality). *There exist two thresholds $\underline{q}_{2c}^V$ and $\underline{q}_{1c}^H$, such that: (1) When $\underline{q}^H < \underline{q}_{1c}^H$ and $\underline{q}^V > \underline{q}_{2c}^V$, virtual avatar streaming actually increases viewers' perceived ELS content quality, i.e., $q_j^{V*} > q_i^{H*}$; (2) When either (i) $\underline{q}^H < \underline{q}_{1c}^H$ and $\underline{q}^V < \underline{q}_{2c}^V$, or (ii) $\underline{q}^H > \underline{q}_{1c}^H$, virtual-avatar streaming actually decreases viewers' perceived ELS content quality, i.e., $q_j^{V*} < q_i^{H*}$.*

(c) (Platform profit). *There exist two thresholds $\underline{q}_{3c}^V$ and $\underline{q}_{2c}^H$, such that: (1) When $\underline{q}^H < \underline{q}_{2c}^H$ and $\underline{q}^V > \underline{q}_{3c}^V$, the platform is actually better off with virtual avatar streaming, i.e., $\bar{R}^{V*} > \bar{R}^{H*}$; (2) When either (i) $\underline{q}^H < \underline{q}_{2c}^H$ and $\underline{q}^V < \underline{q}_{3c}^V$,or (ii) $\underline{q}^H > \underline{q}_{2c}^H$, the platform is actually better off with human streaming, i.e., $\bar{R}^{V*} < \bar{R}^{H*}$.*

(d) (Streamer profit). *There exist two thresholds $\underline{q}_{3c}^H$ and $\underline{q}_{4c}^V$, such that: (1) When $\underline{q}^H > \underline{q}_{3c}^H$ and $\underline{q}^V < \underline{q}_{4c}^V$, the streamer is actually better off from adopting virtual avatar streaming, i.e., $S_j^{V*} > S_i^{H*}$; (2) When either (i) $\underline{q}^H > \underline{q}_{3c}^H$ and $\underline{q}^V > \underline{q}_{4c}^V$, or (ii) $\underline{q}^H < \underline{q}_{3c}^H$, the streamer is actually worse off from adopting virtual avatar streaming, i.e., $S_j^{V*} < S_i^{H*}$.*

Proof.

Part (a).

We have $\alpha^{V*}-\alpha^{H*} = \frac{k\underline{q}^H(1-\eta^H)[1+\eta^H(m-1)]}{2[1+\eta^H(m-2)]} - \frac{\underline{q}^V(1-\eta^V)[1+\eta^V(n-m-1)]}{2\lambda^2[1+\eta^V(n-m-2)]}$. Therefore, $\alpha^{V*} > \alpha^{H*}$ if and only if $\underline{q}^V < \underline{q}_{1c}^V = k\lambda^2\underline{q}^H\frac{(1-\eta^H)[1+\eta^H(m-1)][1+\eta^V(n-m-2)]}{[1+\eta^H(m-2)](1-\eta^V)[1+\eta^V(n-m-1)]}$. We

[a]To isolate the effect of the streaming model from the effect of the number of streamers, we compare the platform's equilibrium profit per streamer (i.e., $\bar{R}^{H*} = \frac{R^{H*}}{m}$ and $\bar{R}^{V*} = \frac{R^{V*}}{n-m}$).

next verify that this threshold always lies within the feasible region. The interior conditions $\alpha^{H*} > 0$ and $\alpha^{V*} > 0$ require $0 < \underline{q}^H < \frac{[\gamma-(1-2\xi)\beta][1+\eta^H(m-2)]}{k(1-\eta^H)[1+\eta^H(m-1)]} = \overline{q}_c^H$ and $0 < \underline{q}^V < \frac{\lambda^2[\gamma-(1-2\xi)\beta][1+\eta^V(n-m-2)]}{(1-\eta^V)[1+\eta^V(n-m-1)]} = \overline{q}_c^V$. It is easy to find that for every feasible $0 < \underline{q}^H < \overline{q}_c^H$, $0 < \underline{q}_{1c}^V < \overline{q}_c^V$. Therefore, no additional threshold for $\underline{q}^H$ is required.

Part (b).

We have $q_j^{V*} - q_i^{H*} = \frac{1}{2}\{\underline{q}^V - \underline{q}^H - (\beta+\gamma)[\frac{1+\eta^H(m-2)}{k(1-\eta^H)[1+\eta^H(m-1)]} - \frac{\lambda^2[1+\eta^V(n-m-2)]}{(1-\eta^V)[1+\eta^V(n-m-1)]}]\}$. Therefore, $q_j^{V*} > q_i^{H*}$ if and only if $\underline{q}^V > \underline{q}_{2c}^V = \underline{q}^H + (\beta+\gamma)[\frac{1+\eta^H(m-2)}{k(1-\eta^H)[1+\eta^H(m-1)]} - \frac{\lambda^2[1+\eta^V(n-m-2)]}{(1-\eta^V)[1+\eta^V(n-m-1)]}]$. Then, the interior condition $\alpha^{V*} > 0$ requires $0 < \underline{q}^V < \overline{q}_c^V$. Substituting the two thresholds and rearranging gives $\underline{q}^H < \underline{q}_{1c}^H = \frac{2\lambda^2(\gamma+\xi\beta)[1+\eta^V(n-m-2)]}{(1-\eta^V)[1+\eta^V(n-m-1)]} - \frac{(\beta+\gamma)[1+\eta^H(m-2)]}{k(1-\eta^H)[1+\eta^H(m-1)]}$. It is easy to find that when $\underline{q}^H < \underline{q}_{1c}^H$, the threshold $\underline{q}_{2c}^V$ lies within the feasible region. It follows that $q_j^{V*} > q_i^{H*}$ when $\underline{q}^V > \underline{q}_{2c}^V$, whereas $q_j^{V*} < q_i^{H*}$ when $\underline{q}^V < \underline{q}_{2c}^V$. By contrast, when $\underline{q}^H > \underline{q}_{1c}^H$, we have $\underline{q}_{2c}^V > \overline{q}_c^V$. Because every feasible $\underline{q}^V$ satisfies $\underline{q}^V < \overline{q}_c^V$, there exists no feasible $\underline{q}^V$ satisfying $\underline{q}^V > \underline{q}_{2c}^V$. Consequently, $q_j^{V*} < q_i^{H*}$ throughout the feasible region.

Part (c).

From Lemma A3, we have $\bar{R}^{H*} = \frac{\{(\beta+\gamma)[1+\eta^H(m-2)]+k\underline{q}^H(1-\eta^H)[1+\eta^H(m-1)]\}^2}{4k(1-\eta^H)[1+\eta^H(m-1)]^2[1+\eta^H(m-2)]}$, and $\bar{R}^{V*} = \frac{\{\lambda^2(\beta+\gamma)[1+\eta^V(n-m-2)]+\underline{q}^V(1-\eta^V)[1+\eta^V(n-m-1)]\}^2}{4\lambda^2(1-\eta^V)[1+\eta^V(n-m-1)]^2[1+\eta^V(n-m-2)]}$. Under the interior-equilibrium conditions, all terms inside the squared expressions are positive. Taking the positive square root of both sides of $\bar{R}^{V*} > \bar{R}^{H*}$ and solving for $\underline{q}^V$ yield $\underline{q}^V > \underline{q}_{3c}^V = \lambda\sqrt{k}\sqrt{\frac{(1-\eta^H)[1+\eta^V(n-m-2)]}{(1-\eta^V)[1+\eta^H(m-2)]}}\underline{q}^H + \frac{\lambda(\beta+\gamma)\sqrt{[1+\eta^H(m-2)][1+\eta^V(n-m-2)]}}{\sqrt{k(1-\eta^H)(1-\eta^V)[1+\eta^H(m-1)]}} - \frac{\lambda^2(\beta+\gamma)[1+\eta^V(n-m-2)]}{(1-\eta^V)[1+\eta^V(n-m-1)]}$. Hence, the platform-profit comparison is characterized by a unique threshold in $\underline{q}^V$. Then, recall that the interior condition $\alpha^{V*} > 0$ requires $0 < \underline{q}^V < \overline{q}_c^V$. Substituting these expressions and rearranging yield $\underline{q}^H < \underline{q}_{2c}^H = \frac{2\lambda(\gamma+\xi\beta)\sqrt{[1+\eta^H(m-2)][1+\eta^V(n-m-2)]}}{\sqrt{k(1-\eta^H)(1-\eta^V)}[1+\eta^V(n-m-1)]} - \frac{(\beta+\gamma)[1+\eta^H(m-2)]}{k(1-\eta^H)[1+\eta^H(m-1)]}$. When $\underline{q}^H < \underline{q}_{2c}^H$, the threshold $\underline{q}_{3c}^V$ lies within the feasible region. Therefore, $\bar{R}^{V*} > \bar{R}^{H*}$ when $\underline{q}^V > \underline{q}_{3c}^V$, whereas $\bar{R}^{V*} < \bar{R}^{H*}$ when $\underline{q}^V < \underline{q}_{3c}^V$. By contrast, when $\underline{q}^H > \underline{q}_{2c}^H$, we have $\underline{q}_{3c}^V > \overline{q}_c^V$. Thus, every feasible $\underline{q}^V$ satisfies $\underline{q}^V < \overline{q}_c^V < \underline{q}_{3c}^V$, which implies $\bar{R}^{V*} < \bar{R}^{H*}$.

Part (d).

For the virtual avatar streamer, we have $\frac{\partial^2 S_j^{V*}}{\partial(\underline{q}^V)^2} = -\frac{3+\eta^V(n-m-4)}{4\lambda^2[1+\eta^V(n-m-2)]} < 0$ and $\frac{\partial S_j^{V*}}{\partial \underline{q}^V} = -\frac{[3+\eta^V(n-m-4)]\underline{q}^V}{4\lambda^2[1+\eta^V(n-m-2)]} + \frac{(\beta+\gamma)[1+\eta^V(n-m-2)]}{4(1-\eta^V)[1+\eta^V(n-m-1)]} - 1$. It is easy to find that $\frac{\partial S_j^{V*}}{\partial \underline{q}^V} < 0$ for every $\underline{q}^V > 0$, and $S_j^{V*}(\underline{q}^V = 0) = \frac{\lambda^2(\beta+\gamma)^2[1+\eta^V(n-m-2)][1-(n-m)\eta^V]}{8(1-\eta^V)^2[1+\eta^V(n-m-1)]^2} > 0$. Similarly, $S_i^{H*}(\underline{q}^H = 0) = \frac{(\beta+\gamma)^2[1+\eta^H(m-2)](1-m\eta^H)}{8k(1-\eta^H)^2[1+\eta^H(m-1)]^2}$. It is easy to find that $S_i^{H*}(\underline{q}^H = 0) > S_j^{V*}(\underline{q}^V = 0) > 0$. Moreover, $\frac{\partial^2 S_i^{H*}}{\partial(\underline{q}^H)^2} = -\frac{k[3+\eta^H(m-4)]}{4[1+\eta^H(m-2)]} < 0$,

whereas $\frac{\partial S_i^{H*}}{\partial \underline{q}^H}\big|_{\underline{q}^H=0} = \frac{(\beta+\gamma)[1+\eta^H(m-2)]}{4(1-\eta^H)[1+\eta^H(m-1)]} > 0$. Thus, S_i^{H*} is strictly concave in $\underline{q}^H$ and initially increases with $\underline{q}^H$. Recall that the interior condition $\alpha^{H*} > 0$ requires $0 < \underline{q}^H < \overline{q}_c^H$. At the upper bound, $e_i^{H*} = 0$ and $\alpha^{H*} = 0$; therefore, $S_i^{H*} = 0 < S_j^{V*}(\underline{q}^V = 0)$. Because S_i^{H*} is continuous and strictly concave, while $S_i^{H*}(\underline{q}^H = 0) > S_j^{V*}(\underline{q}^V = 0)$ and its profit at the upper bound is below $S_j^{V*}(\underline{q}^V = 0)$, there exists a unique threshold $\underline{q}_{3c}^H$ satisfying $S_i^{H*}(\underline{q}_{3c}^H) = S_j^{V*}(\underline{q}^V = 0)$. Consequently, $S_j^{V*}(\underline{q}^V = 0) > S_i^{H*}$ if and only if $\underline{q}^H > \underline{q}_{3c}^H$. When $\underline{q}^H < \underline{q}_{3c}^H$, we have $S_i^{H*} > S_j^{V*}(\underline{q}^V = 0)$. Since S_j^{V*} strictly decreases with $\underline{q}^V$, $S_j^{V*} < S_j^{V*}(\underline{q}^V = 0) < S_i^{H*}$ for every feasible $\underline{q}^V > 0$. Then, when $\underline{q}^H > \underline{q}_{3c}^H$, we have $S_j^{V*}(\underline{q}^V = 0) > S_i^{H*} > 0$. The interior condition $\alpha^{V*} > 0$ requires $0 < \underline{q}^V < \overline{q}_c^V$. At this upper bound, $e_j^{V*} = 0$ and $\alpha^{V*} = 0$, implying $S_j^{V*} = -\underline{q}^V < 0 < S_i^{H*}$. Since S_j^{V*} is continuous and strictly decreasing in $\underline{q}^V$, there exists a unique threshold $\underline{q}_{4c}^V$ satisfying $S_j^{V*} = S_i^{H*}$. Therefore, $S_j^{V*} > S_i^{H*}$ if and only if $\underline{q}^H > \underline{q}_{3c}^H$ and $0 < \underline{q}^V < \underline{q}_{4c}^V$. Otherwise, $S_j^{V*} < S_i^{H*}$. □

F.2. *Disaggregated viewers and network effects*

F.2.1. *Equilibrium results*

Lemma A4.

The equilibrium results are as follows:

$$e^{H*} = \frac{\kappa_N(\beta+\gamma) - k\underline{q}^H}{2k}$$

$$e^{V*} = \frac{\kappa_N\lambda^2(\beta+\gamma) - \underline{q}^V}{2\lambda}$$

$$q^{H*} = \frac{\kappa_N(\beta+\gamma) + k\underline{q}^H}{2k}$$

$$q^{V*} = \frac{\kappa_N\lambda^2(\beta+\gamma) + \underline{q}^V}{2}$$

$$\alpha^{H*} = \frac{\kappa_N[\gamma - (1-2\xi)\beta] - k\underline{q}^H}{2\kappa_N}$$

$$\alpha^{V*} = \frac{\kappa_N\lambda^2[\gamma - (1-2\xi)\beta] - \underline{q}^V}{2\kappa_N\lambda^2}$$

$$S^{H*} = \frac{\left[\kappa_N(\beta+\gamma) - k\underline{q}^H\right]\left[\kappa_N(\beta+\gamma) + 3k\underline{q}^H\right]}{8k}$$

$$S^{V*} = \frac{\left[\kappa_N \lambda^2(\beta+\gamma) - \underline{q}^V\right]\left[\kappa_N \lambda^2(\beta+\gamma) + 3\underline{q}^V\right]}{8\lambda^2} - \underline{q}^V$$

$$R^{H*} = \frac{\left[\kappa_N(\beta+\gamma) + k\underline{q}^H\right]^2}{4k}$$

$$R^{V*} = \frac{\left[\kappa_N \lambda^2(\beta+\gamma) + \underline{q}^V\right]^2}{4\lambda^2}$$

Proof.

Similar to the proof of the main model, we use backward induction. In Stage 3, the F.O.C for viewer r is $\frac{\partial Q_r^o}{\partial \omega_r^o} = q^o - \omega_r^o + \frac{\rho}{N-1} \sum_{s \in \mathcal{N}, s \neq r} \omega_s^o = 0$. In a symmetric equilibrium, $\omega_s^o = \omega_r^o$ for all $s \neq r$, which gives $\omega_r^o = \frac{q^o}{1-\rho}$ and $\Omega^o = \sum_{r=1}^{N} \omega_r^o = \kappa_N q^o$. In Stage 2, substituting $\Omega^H = \kappa_N(\underline{q}^H + e^H)$ and $\Omega^V = \kappa_N(\underline{q}^V + \lambda e^V)$ into the streamer's profit function yields $e^H = \frac{\kappa_N[\alpha^H + (1-\xi)\beta]}{k}$ and $e^V = \kappa_N \lambda[\alpha^V + (1-\xi)\beta]$. In Stage 1, substituting these effort responses into the platform's profit functions and taking the F.O.C with respect to α^H and α^V yields $\alpha^{H*} = \frac{\kappa_N[\gamma - (1-2\xi)\beta] - k\underline{q}^H}{2\kappa_N}$ and $\alpha^{V*} = \frac{\kappa_N \lambda^2[\gamma - (1-2\xi)\beta] - \underline{q}^V}{2\kappa_N \lambda^2}$. The relevant second-order derivatives are negative, i.e., $\frac{\partial^2 Q_r^o}{\partial(\omega_r^o)^2} = -1$, $\frac{\partial^2 S^H}{\partial(e^H)^2} = -k$, $\frac{\partial^2 S^V}{\partial(e^V)^2} = -1$, $\frac{\partial^2 R^H}{\partial(\alpha^H)^2} = -\frac{2\kappa_N^2}{k}$, and $\frac{\partial^2 R^V}{\partial(\alpha^V)^2} = -2\kappa_N^2 \lambda^2$. Substituting the equilibrium creative rewards into the effort responses and then into the content quality, content consumption, and profit functions gives the stated results.

It is worth noting that the following sufficient conditions are imposed to ensure that $0 < \alpha^{H*} < \gamma$ and $0 < \alpha^{V*} < \gamma$. (1) $\gamma > |1 - 2\xi|\beta$, (2) $0 < \underline{q}^H < \frac{\kappa_N[\gamma - (1-2\xi)\beta]}{k}$, and (3) $0 < \underline{q}^V < \kappa_N \lambda^2[\gamma - (1-2\xi)\beta]$. □

F.2.2. *Robustness test*

We next examine whether the main findings in Propositions 1–4 remain valid in this setting.

Proposition A2.

(a) (Creative reward). *There exists a threshold $\underline{q}_{1N}^V$ such that: (1) When $\underline{q}^V < \underline{q}_{1N}^V$, the platform tends to offer a higher creative reward to virtual avatar streamers, i.e., $\alpha^{V*} > \alpha^{H*}$; (2) When $\underline{q}^V > \underline{q}_{1N}^V$, the platform tends to offer a lower creative reward to virtual avatar streamers, i.e., $\alpha^{V*} < \alpha^{H*}$.*

(b) (Viewers' perceived ELS content quality). *There exist two thresholds $\underline{q}_{2N}^V$ and $\underline{q}_{1N}^H$, such that: (1) When $\underline{q}^H < \underline{q}_{1N}^H$ and $\underline{q}^V > \underline{q}_{2N}^V$, virtual avatar streaming actually increases viewers' perceived ELS content quality, i.e., $q^{V*} > q^{H*}$; (2) When*

either (i) $\underline{q}^H < \underline{q}^H_{1N}$ *and* $\underline{q}^V < \underline{q}^V_{2N}$*, or (ii)* $\underline{q}^H > \underline{q}^H_{1N}$*, virtual avatar streaming actually decreases viewers' perceived ELS content quality, i.e.,* $q^{V*} < q^{H*}$*.*

(c) (Platform profit). *There exist two thresholds* $\underline{q}^V_{3N}$ *and* $\underline{q}^H_{2N}$*, such that: (1) When* $\underline{q}^H < \underline{q}^H_{2N}$ *and* $\underline{q}^V > \underline{q}^V_{3N}$*, the platform is actually better off with virtual avatar streaming, i.e.,* $R^{V*} > R^{H*}$*; (2) When either (i)* $\underline{q}^H < \underline{q}^H_{2N}$ *and* $\underline{q}^V < \underline{q}^V_{3N}$*, or (ii)* $\underline{q}^H > \underline{q}^H_{2N}$*, the platform is actually better off with human streaming, i.e.,* $R^{V*} < R^{H*}$*.*

(d) (Streamer profit). *There exist two thresholds* $\underline{q}^H_{3N}$ *and* $\underline{q}^V_{4N}$*, such that: (1) When* $\underline{q}^H > \underline{q}^H_{3N}$ *and* $\underline{q}^V < \underline{q}^V_{4N}$*, the streamer is actually better off from adopting virtual avatar streaming, i.e.,* $S^{V*} > S^{H*}$*; (2) When either (i)* $\underline{q}^H > \underline{q}^H_{3N}$ *and* $\underline{q}^V > \underline{q}^V_{4N}$*, or (ii)* $\underline{q}^H < \underline{q}^H_{3N}$*, the streamer is actually worse off from adopting virtual avatar streaming, i.e.,* $S^{V*} < S^{H*}$*.*

Proof.

Part (a).

From Lemma A4, we have $\alpha^{V*} - \alpha^{H*} = \frac{k\lambda^2 \underline{q}^H - \underline{q}^V}{2\kappa_N \lambda^2}$. Therefore, $\alpha^{V*} > \alpha^{H*}$ if and only if $\underline{q}^V < \underline{q}^V_{1N} = k\lambda^2 \underline{q}^H$. The interior conditions imply $0 < \underline{q}^H < \frac{\kappa_N[\gamma-(1-2\xi)\beta]}{k}$ and $0 < \underline{q}^V < \kappa_N\lambda^2[\gamma-(1-2\xi)\beta]$. Hence, $\underline{q}^V_{1N}$ always lies within the feasible region.

Part (b).

We have $q^{V*} - q^{H*} = \frac{1}{2}\left[\underline{q}^V - \underline{q}^H - \kappa_N(\beta+\gamma)\left(\frac{1}{k} - \lambda^2\right)\right]$. Thus, $q^{V*} > q^{H*}$ if and only if $\underline{q}^V > \underline{q}^V_{2N} = \underline{q}^H + \kappa_N(\beta+\gamma)\left(\frac{1}{k} - \lambda^2\right)$. Comparing $\underline{q}^V_{2N}$ with the upper feasible bound of $\underline{q}^V$ gives $\underline{q}^H < \underline{q}^H_{1N} = \kappa_N\left[2\lambda^2(\gamma + \xi\beta) - \frac{\beta+\gamma}{k}\right]$. Therefore, when $\underline{q}^H < \underline{q}^H_{1N}$, the threshold $\underline{q}^V_{2N}$ lies within the feasible region and determines the content-quality comparison. When $\underline{q}^H > \underline{q}^H_{1N}$, $\underline{q}^V_{2N}$ lies above the feasible upper bound, so that $q^{V*} < q^{H*}$ throughout the feasible region.

Part (c).

Because the terms inside the squared expressions in R^{H*} and R^{V*} are positive, $R^{V*} > R^{H*}$ is equivalent to $\frac{\kappa_N\lambda^2(\beta+\gamma)+\underline{q}^V}{\lambda} > \frac{\kappa_N(\beta+\gamma)+k\underline{q}^H}{\sqrt{k}}$. Solving this inequality gives $\underline{q}^V > \underline{q}^V_{3N} = \lambda\sqrt{k}\,\underline{q}^H + \kappa_N(\beta+\gamma)\left(\frac{\lambda}{\sqrt{k}} - \lambda^2\right)$. The threshold $\underline{q}^V_{3N}$ lies below the feasible upper bound of $\underline{q}^V$ if and only if $\underline{q}^H < \underline{q}^H_{2N} = \kappa_N\left[\frac{2\lambda(\gamma+\xi\beta)}{\sqrt{k}} - \frac{\beta+\gamma}{k}\right]$. Hence, when $\underline{q}^H < \underline{q}^H_{2N}$, $\underline{q}^V_{3N}$ determines the platform-profit comparison. When $\underline{q}^H > \underline{q}^H_{2N}$, the threshold exceeds the feasible upper bound and $R^{V*} < R^{H*}$ throughout the feasible region.

Part (d).

From Lemma A4, we have $S^{H*} = \frac{[\kappa_N(\beta+\gamma)-k\underline{q}^H][\kappa_N(\beta+\gamma)+3k\underline{q}^H]}{8k}$ and $S^{V*} = \frac{[\kappa_N\lambda^2(\beta+\gamma)-\underline{q}^V][\kappa_N\lambda^2(\beta+\gamma)+3\underline{q}^V]}{8\lambda^2} - \underline{q}^V$. Moreover, $\frac{\partial S^{V*}}{\partial \underline{q}^V} = \frac{\kappa_N(\beta+\gamma)}{4} - 1 - \frac{3\underline{q}^V}{4\lambda^2}$. It is easy to find that S^{V*} is strictly decreasing in $\underline{q}^V > 0$. Then, we have $S^{H*}(\underline{q}^H = 0) - S^{V*}(\underline{q}^V = 0) = \frac{\kappa_N^2(\beta+\gamma)^2}{8}\left(\frac{1}{k} - \lambda^2\right) > 0$ under $k\lambda^2 < 1$. Solving $S^{H*} = S^{V*}(\underline{q}^V = 0)$ gives the unique positive threshold $\underline{q}^H_{3N} = \frac{\kappa_N(\beta+\gamma)}{3k}\left(1 + \sqrt{4 - 3k\lambda^2}\right)$.

When $\underline{q}^H < \underline{q}^H_{3N}$, $S^{H*} > S^{V*}(\underline{q}^V = 0)$, and the monotonicity of S^{V*} implies $S^{H*} > S^{V*}$ for every feasible $\underline{q}^V > 0$. When $\underline{q}^H > \underline{q}^H_{3N}$, $S^{V*}(\underline{q}^V = 0) > S^{H*}$. By continuity and strict monotonicity, there exists a unique threshold $\underline{q}^V_{4N}$ satisfying $S^{V*}(\underline{q}^V_{4N}) = S^{H*}$. Therefore, $S^{V*} > S^{H*}$ when $\underline{q}^V < \underline{q}^V_{4N}$, whereas $S^{V*} < S^{H*}$ when $\underline{q}^V > \underline{q}^V_{4N}$. □

F.3. *Heterogeneous viewer preferences*

F.3.1. *Equilibrium results*

Lemma A5.

Let $A^H(q^H, q^V) = \frac{\partial \Omega^H(q^H, q^V)}{\partial q^H} = 1 + \frac{3(q^H)^2 - (q^V)^2}{2t}$ *and* $A^V(q^H, q^V) = \frac{\partial \Omega^V(q^H, q^V)}{\partial q^V} = 1 + \frac{3(q^V)^2 - (q^H)^2}{2t}$. *The equilibrium perceived ELS content qualities* (q^{H*}, q^{V*}) *satisfy* $(\beta+\gamma)A^H - k(q^{H*} - \underline{q}^H) - \frac{k[A^H - \frac{3q^{H*}(q^{H*} - \underline{q}^H)}{t}]}{(A^H)^2}\Omega^{H*} - \frac{q^{H*}q^{V*}}{t}[\beta + \gamma - \frac{q^{V*} - \underline{q}^V}{\lambda^2 A^V}] - \frac{q^{H*}(q^{V*} - \underline{q}^V)}{\lambda^2 t (A^V)^2}\Omega^{V*} = 0$ *and* $(\beta+\gamma)A^V - \frac{q^{V*} - \underline{q}^V}{\lambda^2} - \frac{A^V - \frac{3q^{V*}(q^{V*} - \underline{q}^V)}{t}}{\lambda^2 (A^V)^2}\Omega^{V*} - \frac{q^{H*}q^{V*}}{t}[\beta + \gamma - \frac{k(q^{H*} - \underline{q}^H)}{A^H}] - \frac{kq^{V*}(q^{H*} - \underline{q}^H)}{t(A^H)^2}\Omega^{H*} = 0$. *On this basis,* $e^{H*} = q^{H*} - \underline{q}^H$, $e^{V*} = \frac{q^{V*} - \underline{q}^V}{\lambda}$, $\alpha^{H*} = \frac{k(q^{H*} - \underline{q}^H)}{A^H} - (1 - \xi)\beta$, $\alpha^{V*} = \frac{q^{V*} - \underline{q}^V}{\lambda^2 A^V} - (1 - \xi)\beta$, $S^{H*} = \frac{k(q^{H*} - \underline{q}^H)}{A^H}\Omega^{H*} - \frac{k(q^{H*} - \underline{q}^H)^2}{2}$, $S^{V*} = \frac{q^{V*} - \underline{q}^V}{\lambda^2 A^V}\Omega^{V*} - \frac{(q^{V*} - \underline{q}^V)^2}{2\lambda^2} - \underline{q}^V$, $R^{H*} = [\beta + \gamma - \frac{k(q^{H*} - \underline{q}^H)}{A^H}]\Omega^{H*}$, $R^{V*} = [\beta + \gamma - \frac{q^{V*} - \underline{q}^V}{\lambda^2 A^V}]\Omega^{V*}$.

Proof.

Similar to the proof of the main model, we use backward induction. In Stage 3, viewers choose between human and virtual avatar streaming and then decide content consumption, yielding $\Omega^H(q^H, q^V)$ and $\Omega^V(q^H, q^V)$. By construction, $\frac{\partial \Omega^H}{\partial q^H} = A^H(q^H, q^V)$, $\frac{\partial \Omega^V}{\partial q^V} = A^V(q^H, q^V)$, and $\frac{\partial \Omega^H}{\partial q^V} = \frac{\partial \Omega^V}{\partial q^H} = -\frac{q^H q^V}{t}$. In Stage 2, the human streamer's F.O.C is $[\alpha^H + (1 - \xi)\beta]A^H(q^H, q^V) - ke^H = 0$, whereas the virtual avatar streamer's F.O.C is $\lambda[\alpha^V + (1 - \xi)\beta]A^V(q^H, q^V) - e^V = 0$. Using $q^H = \underline{q}^H + e^H$ and $q^V = \underline{q}^V + \lambda e^V$, these conditions yield $\alpha^H = \frac{k(q^H - \underline{q}^H)}{A^H(q^H, q^V)} - (1 - \xi)\beta$ and $\alpha^V = \frac{q^V - \underline{q}^V}{\lambda^2 A^V(q^H, q^V)} - (1 - \xi)\beta$. In Stage 1, substituting the streamers' optimal responses into the platform's profit function gives $R = [\beta + \gamma - \frac{k(q^H - \underline{q}^H)}{A^H(q^H, q^V)}]\Omega^H(q^H, q^V) + [\beta + \gamma - \frac{q^V - \underline{q}^V}{\lambda^2 A^V(q^H, q^V)}]\Omega^V(q^H, q^V)$. Differentiating this expression with respect to q^H and q^V yields the two equilibrium equations stated in Lemma A5. Substituting their unique interior solution into the effort, creative-reward, aggregate-consumption, and profit functions gives the remaining equilibrium results.

It is worth noting that an interior equilibrium requires $0 < \alpha^{H*} < \gamma$, $0 < \alpha^{V*} < \gamma$, $|(q^{H*})^2 - (q^{V*})^2| < 2t$, $A^H > \frac{3q^{H*}(q^{H*} - \underline{q}^H)}{t}$, and $A^V > \frac{3q^{V*}(q^{V*} - \underline{q}^V)}{t}$, together with the platform's second-order conditions. □

F.3.2. *Robustness test*

We next examine whether the main findings in Propositions 1–4 remain valid in this setting.

Proposition A3. *Suppose that $\chi^* \leq \varepsilon < 1$, where $\varepsilon > 0$ is sufficiently small. Then:*

(a) (Creative reward). *There exists a threshold $\underline{q}^V_{1x}$ such that: (1) When $\underline{q}^V < \underline{q}^V_{1x}$, the platform tends to offer a higher creative reward to the virtual avatar streamer, i.e., $\alpha^{V*} > \alpha^{H*}$; (2) When $\underline{q}^V > \underline{q}^V_{1x}$, the platform tends to offer a lower creative reward to the virtual avatar streamer, i.e., $\alpha^{V*} < \alpha^{H*}$.*

(b) (Viewers' perceived ELS content quality). *There exist two thresholds $\underline{q}^V_{2x}$ and $\underline{q}^H_{1x}$, such that: (1) When $\underline{q}^H < \underline{q}^H_{1x}$ and $\underline{q}^V > \underline{q}^V_{2x}$, virtual avatar streaming actually increases viewers' perceived ELS content quality, i.e., $q^{V*} > q^{H*}$; (2) When either (i) $\underline{q}^H < \underline{q}^H_{1x}$ and $\underline{q}^V < \underline{q}^V_{2x}$, or (ii) $\underline{q}^H > \underline{q}^H_{1x}$, virtual avatar streaming actually decreases viewers' perceived ELS content quality, i.e., $q^{V*} < q^{H*}$.*

(c) (Platform profit). *There exist two thresholds $\underline{q}^V_{3x}$ and $\underline{q}^H_{2x}$, such that: (1) When $\underline{q}^H < \underline{q}^H_{2x}$ and $\underline{q}^V > \underline{q}^V_{3x}$, the platform is actually better off with virtual avatar streaming, i.e., $R^{V*} > R^{H*}$; (2) When either (i) $\underline{q}^H < \underline{q}^H_{2x}$ and $\underline{q}^V < \underline{q}^V_{3x}$, or (ii) $\underline{q}^H > \underline{q}^H_{2x}$, the platform is actually better off with human streaming, i.e., $R^{V*} < R^{H*}$.*

(d) (Streamer profit). *There exist two thresholds $\underline{q}^H_{3x}$ and $\underline{q}^V_{4x}$, such that: (1) When $\underline{q}^H > \underline{q}^H_{3x}$ and $\underline{q}^V < \underline{q}^V_{4x}$, the streamer is actually better off from adopting virtual avatar streaming, i.e., $S^{V*} > S^{H*}$; (2) When either (i) $\underline{q}^H > \underline{q}^H_{3x}$ and $\underline{q}^V > \underline{q}^V_{4x}$, or (ii) $\underline{q}^H < \underline{q}^H_{3x}$, the streamer is actually worse off from adopting virtual avatar streaming, i.e., $S^{V*} < S^{H*}$.*

Proof.

According to Remark 1, $\frac{|\Omega^{H*} - q^{H*}|}{q^{H*}} \leq \varepsilon$ and $\frac{|\Omega^{V*} - q^{V*}|}{q^{V*}} \leq \varepsilon$. Moreover, $|A^H - 1| \leq 3\varepsilon$, $|A^V - 1| \leq 3\varepsilon$, and $\frac{q^{H*} q^{V*}}{t} \leq \varepsilon$. These bounds imply that the additional terms arising from endogenous viewer choices are of order $O(\varepsilon)$. Applying these bounds to the two exact equilibrium equations in Lemma A5 yields $\beta + \gamma + k\underline{q}^H - 2kq^{H*} = O(\varepsilon)$ and $\lambda^2(\beta + \gamma) + \underline{q}^V - 2q^{V*} = O(\varepsilon)$. It follows that $q^{H*} = \frac{\beta + \gamma + k\underline{q}^H}{2k} + O(\varepsilon)$ and $q^{V*} = \frac{\lambda^2(\beta+\gamma) + \underline{q}^V}{2} + O(\varepsilon)$.

Part (a).

From Lemma A5, we have $\alpha^{V*} - \alpha^{H*} = \frac{k\lambda^2 \underline{q}^H - \underline{q}^V}{2\lambda^2} + O(\varepsilon)$ and $\frac{\partial(\alpha^{V*} - \alpha^{H*})}{\partial \underline{q}^V} = -\frac{1}{2\lambda^2} + O(\varepsilon) < 0$. It follows that there exists a unique threshold $\underline{q}^V_{1x} = k\lambda^2 \underline{q}^H + O(\varepsilon)$ such that $\alpha^{V*} > \alpha^{H*}$ if and only if $\underline{q}^V < \underline{q}^V_{1x}$. The reward-feasibility conditions imply $\underline{q}^H < \frac{\gamma - (1-2\xi)\beta}{k} + O(\varepsilon)$ and $\underline{q}^V < \lambda^2[\gamma - (1-2\xi)\beta] + O(\varepsilon)$. Consequently, the threshold $\underline{q}^V_{1x}$ remains inside the perturbed feasible region whenever the corresponding main-model threshold lies strictly inside its feasible region.

Part (b).

Similar to the above, we have $q^{V*} - q^{H*} = \frac{1}{2}[\underline{q}^V - \underline{q}^H - (\beta + \gamma)(\frac{1}{k} - \lambda^2)] + O(\varepsilon)$ and

$\frac{\partial(q^{V*}-q^{H*})}{\partial \underline{q}^V} = \frac{1}{2} + O(\varepsilon) > 0$. Therefore, there exists a unique threshold $\underline{q}^V_{2x} = \underline{q}^H + (\beta+\gamma)(\frac{1}{k} - \lambda^2) + O(\varepsilon)$ such that $q^{V*} > q^{H*}$ if and only if $\underline{q}^V > \underline{q}^V_{2x}$. Comparing this threshold with the feasible upper bound of $\underline{q}^V$ gives $\underline{q}^H_{1x} = 2\lambda^2(\gamma+\xi\beta) - \frac{\beta+\gamma}{k} + O(\varepsilon)$. When $\underline{q}^H < \underline{q}^H_{1x}$, the threshold $\underline{q}^V_{2x}$ lies inside the feasible region, so the quality ranking changes at $\underline{q}^V_{2x}$. When $\underline{q}^H > \underline{q}^H_{1x}$, the threshold lies above the feasible range of $\underline{q}^V$, implying $q^{V*} < q^{H*}$ throughout the feasible region.

Part (c).

We have $R^{V*} - R^{H*} = \frac{[\lambda^2(\beta+\gamma)+\underline{q}^V]^2}{4\lambda^2} - \frac{[\beta+\gamma+k\underline{q}^H]^2}{4k} + O(\varepsilon)$ and $\frac{\partial(R^{V*}-R^{H*})}{\partial \underline{q}^V} = \frac{\lambda^2(\beta+\gamma)+\underline{q}^V}{2\lambda^2} + O(\varepsilon) > 0$. It follows that there exists a unique threshold $\underline{q}^V_{3x} = \lambda\sqrt{k}\underline{q}^H + (\beta+\gamma)(\frac{\lambda}{\sqrt{k}} - \lambda^2) + O(\varepsilon)$ such that $R^{V*} > R^{H*}$ if and only if $\underline{q}^V > \underline{q}^V_{3x}$. Comparing this threshold with the feasible upper bound of $\underline{q}^V$ gives $\underline{q}^H_{2x} = \frac{2\lambda(\gamma+\xi\beta)}{\sqrt{k}} - \frac{\beta+\gamma}{k} + O(\varepsilon)$. When $\underline{q}^H < \underline{q}^H_{2x}$, $\underline{q}^V_{3x}$ lies inside the feasible region and determines the platform-profit comparison. When $\underline{q}^H > \underline{q}^H_{2x}$, the threshold lies above the feasible range of $\underline{q}^V$, implying $R^{V*} < R^{H*}$ throughout the feasible region.

Part (d).

The equilibrium streamer profits satisfy $S^{H*} = \frac{[\beta+\gamma-k\underline{q}^H][\beta+\gamma+3k\underline{q}^H]}{8k} + O(\varepsilon)$ and $S^{V*} = \frac{[\lambda^2(\beta+\gamma)-\underline{q}^V][\lambda^2(\beta+\gamma)+3\underline{q}^V]}{8\lambda^2} - \underline{q}^V + O(\varepsilon)$. At $\underline{q}^V = 0$, their difference is $S^{V*}(\underline{q}^V = 0) - S^{H*} = \frac{\lambda^2(\beta+\gamma)^2}{8} - \frac{[\beta+\gamma-k\underline{q}^H][\beta+\gamma+3k\underline{q}^H]}{8k} + O(\varepsilon)$. By continuity of the equilibrium around the main-model solution, there exists a threshold $\underline{q}^H_{3x} = \frac{\beta+\gamma}{3k}[1+\sqrt{4-3k\lambda^2}] + O(\varepsilon)$ such that $S^{V*}(\underline{q}^V = 0) > S^{H*}$ if and only if $\underline{q}^H > \underline{q}^H_{3x}$. Moreover, $\frac{\partial S^{V*}}{\partial \underline{q}^V} = \frac{\beta+\gamma}{4} - 1 - \frac{3\underline{q}^V}{4\lambda^2} + O(\varepsilon) < 0$ under the parameter restrictions maintained in the main model. When $\underline{q}^H < \underline{q}^H_{3x}$, $S^{V*}(\underline{q}^V = 0) < S^{H*}$ and therefore $S^{V*} < S^{H*}$ for every feasible $\underline{q}^V > 0$. When $\underline{q}^H > \underline{q}^H_{3x}$, $S^{V*}(\underline{q}^V = 0) > S^{H*}$, and continuity implies the existence of a unique threshold $\underline{q}^V_{4x} = \underline{q}^V_4 + O(\varepsilon)$ satisfying $S^{V*}(\underline{q}^V_{4x}) = S^{H*}$. Therefore, $S^{V*} > S^{H*}$ when $\underline{q}^H > \underline{q}^H_{3x}$ and $\underline{q}^V < \underline{q}^V_{4x}$, whereas $S^{V*} < S^{H*}$ otherwise. □

F.4. *Nonmonetary incentives*

F.4.1. *Equilibrium results*

Lemma A6.

The equilibrium results are as follows:

$$e^{H*} = \frac{\mu[\beta+\gamma-k\underline{q}^H] + (\beta+\gamma)^2\underline{q}^H}{2k\mu - (\beta+\gamma)^2}$$

$$e^{V*} = \frac{\mu[\lambda^2(\beta+\gamma) - \underline{q}^V] + \lambda^2(\beta+\gamma)^2\underline{q}^V}{\lambda[2\mu - \lambda^2(\beta+\gamma)^2]}$$

$$q^{H*} = \frac{\mu[\beta + \gamma + k\underline{q}^H]}{2k\mu - (\beta+\gamma)^2}$$

$$q^{V*} = \frac{\mu[\underline{q}^V + \lambda^2(\beta+\gamma)]}{2\mu - \lambda^2(\beta+\gamma)^2}$$

$$\alpha^{H*} = \frac{\mu[\gamma - (1-2\xi)\beta] - \underline{q}^H[k\mu - (\beta+\gamma)(\gamma+\xi\beta)]}{2\mu + (\beta+\gamma)\underline{q}^H}$$

$$\alpha^{V*} = \frac{\mu\lambda^2[\gamma - (1-2\xi)\beta] - \underline{q}^V[\mu - \lambda^2(\beta+\gamma)(\gamma+\xi\beta)]}{\lambda^2[2\mu + (\beta+\gamma)\underline{q}^V]}$$

$$I^{H*} = \frac{(\beta+\gamma)[\beta + \gamma + k\underline{q}^H]}{2k\mu - (\beta+\gamma)^2}$$

$$I^{V*} = \frac{(\beta+\gamma)[\underline{q}^V + \lambda^2(\beta+\gamma)]}{2\mu - \lambda^2(\beta+\gamma)^2}$$

$$S^{H*} = \frac{k[\mu[\beta+\gamma-k\underline{q}^H] + (\beta+\gamma)^2\underline{q}^H][\mu[\beta+\gamma+3k\underline{q}^H] - (\beta+\gamma)^2\underline{q}^H]}{2[2k\mu - (\beta+\gamma)^2]^2}$$

$$S^{V*} = \frac{[\mu[\lambda^2(\beta+\gamma) - \underline{q}^V] + \lambda^2(\beta+\gamma)^2\underline{q}^V][\mu[\lambda^2(\beta+\gamma) + 3\underline{q}^V] - \lambda^2(\beta+\gamma)^2\underline{q}^V]}{2\lambda^2[2\mu - \lambda^2(\beta+\gamma)^2]^2} - \underline{q}^V$$

$$R^{H*} = \frac{\mu[\beta + \gamma + k\underline{q}^H]^2}{2[2k\mu - (\beta+\gamma)^2]}$$

$$R^{V*} = \frac{\mu[\underline{q}^V + \lambda^2(\beta+\gamma)]^2}{2\lambda^2[2\mu - \lambda^2(\beta+\gamma)^2]}$$

Proof.

Similar to the proof of the main model, we use backward induction. In Stage 3, taking the F.O.C. of Q^o with respect to ω^o gives $\frac{\partial Q^o}{\partial \omega^o} = (1+I^o)q^o - \omega^o = 0$, and hence $\omega^o = (1+I^o)q^o$. Moreover, $\frac{\partial^2 Q^o}{\partial(\omega^o)^2} = -1 < 0$ ensures that this content-consumption decision is optimal. In Stage 2, substituting $\omega^H = (1+I^H)(\underline{q}^H + e^H)$ and $\omega^V = (1+I^V)(\underline{q}^V + \lambda e^V)$ into the streamers' profit functions gives $e^H = \frac{(1+I^H)[\alpha^H + (1-\xi)\beta]}{k}$ and $e^V = \lambda(1+I^V)[\alpha^V + (1-\xi)\beta]$, where $\frac{\partial^2 S^H}{\partial(e^H)^2} = -k < 0$ and $\frac{\partial^2 S^V}{\partial(e^V)^2} = -1 < 0$. In Stage 1, Models H and V are solved separately. Under Model H, substituting the effort response into the platform's profit function and jointly

optimizing with respect to α^H and I^H yields F.O.C.s that can be simplified to $\alpha^H = \frac{\gamma-(1-2\xi)\beta}{2} - \frac{k\underline{q}^H}{2(1+I^H)}$ and $\mu I^H = \frac{\beta+\gamma}{2k}[(\beta+\gamma)(1+I^H)+k\underline{q}^H]$. For any given I^H, $\frac{\partial^2 R^H}{\partial(\alpha^H)^2} = -\frac{2(1+I^H)^2}{k} < 0$. Substituting the optimal creative-reward response into the platform's profit function gives the reduced profit $R^H(I^H) = \frac{[(\beta+\gamma)(1+I^H)+k\underline{q}^H]^2}{4k} - \frac{\mu(I^H)^2}{2}$, with $\frac{\partial^2 R^H}{\partial(I^H)^2} = \frac{(\beta+\gamma)^2}{2k} - \mu < 0$. Similarly, under Model V, the F.O.C.s can be simplified to $\alpha^V = \frac{\gamma-(1-2\xi)\beta}{2} - \frac{\underline{q}^V}{2\lambda^2(1+I^V)}$ and $\mu I^V = \frac{\beta+\gamma}{2}[\lambda^2(\beta+\gamma)(1+I^V)+\underline{q}^V]$. For any given I^V, $\frac{\partial^2 R^V}{\partial(\alpha^V)^2} = -2\lambda^2(1+I^V)^2 < 0$. Substituting the optimal creative-reward response gives the reduced profit $R^V(I^V) = \frac{[\lambda^2(\beta+\gamma)(1+I^V)+\underline{q}^V]^2}{4\lambda^2} - \frac{\mu(I^V)^2}{2}$, with $\frac{\partial^2 R^V}{\partial(I^V)^2} = \frac{\lambda^2(\beta+\gamma)^2}{2} - \mu < 0$. Solving the corresponding F.O.C.s and substituting the equilibrium creative rewards and nonmonetary support levels into the effort, perceived-quality, and payoff functions gives the stated results.

It is worth noting that the following sufficient conditions are imposed to ensure that $0 < \alpha^{H*} < \gamma$, $0 < \alpha^{V*} < \gamma$, $I^{H*} > 0$, and $I^{V*} > 0$. (1) $\gamma > |1-2\xi|\beta$; (2) $\mu > \max\{\frac{(\beta+\gamma)(\gamma+\xi\beta)}{k}, \lambda^2(\beta+\gamma)(\gamma+\xi\beta)\}$; (3) $0 < \underline{q}^H < \frac{\mu[\gamma-(1-2\xi)\beta]}{k\mu-(\beta+\gamma)(\gamma+\xi\beta)}$; and (4) $0 < \underline{q}^V < \frac{\mu\lambda^2[\gamma-(1-2\xi)\beta]}{\mu-\lambda^2(\beta+\gamma)(\gamma+\xi\beta)}$. □

F.4.2. *Robustness test*

We next examine whether the main findings in Propositions 1–4 remain valid in this setting.

Proposition A4.

(a) (Creative reward). *There exists a threshold $\underline{q}^V_{1t}$ such that: (1) When $\underline{q}^V < \underline{q}^V_{1t}$, the platform tends to offer a higher creative reward to the virtual avatar streamer, i.e., $\alpha^{V*} > \alpha^{H*}$; (2) When $\underline{q}^V > \underline{q}^V_{1t}$, the platform tends to offer a lower creative reward to the virtual avatar streamer, i.e., $\alpha^{V*} < \alpha^{H*}$.*

(b) (Viewers' perceived ELS content quality). *There exist two thresholds $\underline{q}^V_{2t}$ and $\underline{q}^H_{1t}$, such that: (1) When $\underline{q}^H < \underline{q}^H_{1t}$ and $\underline{q}^V > \underline{q}^V_{2t}$, virtual avatar streaming actually increases viewers' perceived ELS content quality, i.e., $q^{V*} > q^{H*}$; (2) When either (i) $\underline{q}^H < \underline{q}^H_{1t}$ and $\underline{q}^V < \underline{q}^V_{2t}$, or (ii) $\underline{q}^H > \underline{q}^H_{1t}$, virtual avatar streaming actually decreases viewers' perceived ELS content quality, i.e., $q^{V*} < q^{H*}$.*

(c) (Platform profit). *There exist two thresholds $\underline{q}^V_{3t}$ and $\underline{q}^H_{2t}$, such that: (1) When $\underline{q}^H < \underline{q}^H_{2t}$ and $\underline{q}^V > \underline{q}^V_{3t}$, the platform is actually better off with virtual avatar streaming, i.e., $R^{V*} > R^{H*}$; (2) When either (i) $\underline{q}^H < \underline{q}^H_{2t}$ and $\underline{q}^V < \underline{q}^V_{3t}$, or (ii) $\underline{q}^H > \underline{q}^H_{2t}$, the platform is actually better off with human streaming, i.e., $R^{V*} < R^{H*}$.*

(d) (Streamer profit). *There exist two thresholds $\underline{q}^H_{3t}$ and $\underline{q}^V_{4t}$, such that: (1) When $\underline{q}^H > \underline{q}^H_{3t}$ and $\underline{q}^V < \underline{q}^V_{4t}$, the streamer is actually better off from adopting virtual avatar streaming, i.e., $S^{V*} > S^{H*}$; (2) When either (i) $\underline{q}^H > \underline{q}^H_{3t}$ and $\underline{q}^V > \underline{q}^V_{4t}$, or (ii) $\underline{q}^H < \underline{q}^H_{3t}$, the streamer is actually worse off from adopting virtual avatar streaming, i.e., $S^{V*} < S^{H*}$.*

Proof.

Part (a).

We have $\alpha^{V*} - \alpha^{H*} = \frac{\mu\{\lambda^2 \underline{q}^H[2k\mu-(\beta+\gamma)^2]-\underline{q}^V[2\mu-\lambda^2(\beta+\gamma)^2+(\beta+\gamma)\underline{q}^H(1-k\lambda^2)]\}}{\lambda^2[2\mu+(\beta+\gamma)\underline{q}^H][2\mu+(\beta+\gamma)\underline{q}^V]}$. Because the denominator is positive, $\alpha^{V*} > \alpha^{H*}$ if and only if $\underline{q}^V < \underline{q}^V_{1t} = \frac{\lambda^2 \underline{q}^H[2k\mu-(\beta+\gamma)^2]}{2\mu-\lambda^2(\beta+\gamma)^2+(\beta+\gamma)\underline{q}^H(1-k\lambda^2)}$. The interior conditions imply $0 < \underline{q}^H < \frac{\mu[\gamma-(1-2\xi)\beta]}{k\mu-(\beta+\gamma)(\gamma+\xi\beta)} = \overline{q}^H_t$ and $0 < \underline{q}^V < \frac{\mu\lambda^2[\gamma-(1-2\xi)\beta]}{\mu-\lambda^2(\beta+\gamma)(\gamma+\xi\beta)} = \overline{q}^V_t$. Moreover, $\underline{q}^V_{1t}$ is strictly increasing in $\underline{q}^H$ and satisfies $\underline{q}^V_{1t}(\overline{q}^H_t) = \overline{q}^V_t$. Therefore, $\underline{q}^V_{1t}$ lies within the feasible region for every feasible $\underline{q}^H$. It follows that $\alpha^{V*} > \alpha^{H*}$ when $\underline{q}^V < \underline{q}^V_{1t}$, whereas $\alpha^{V*} < \alpha^{H*}$ when $\underline{q}^V > \underline{q}^V_{1t}$.

Part (b).

Similar to the above, we have $q^{V*} - q^{H*} = \frac{\mu}{2\mu-\lambda^2(\beta+\gamma)^2}[\underline{q}^V - \underline{q}^V_{2t}]$, where $\underline{q}^V_{2t} = \frac{[2\mu-\lambda^2(\beta+\gamma)^2][\beta+\gamma+k\underline{q}^H]}{2k\mu-(\beta+\gamma)^2} - \lambda^2(\beta+\gamma)$. Therefore, $q^{V*} > q^{H*}$ if and only if $\underline{q}^V > \underline{q}^V_{2t}$. The feasible upper bound of $\underline{q}^V$ is $\overline{q}^V_t = \frac{\mu\lambda^2[\gamma-(1-2\xi)\beta]}{\mu-\lambda^2(\beta+\gamma)(\gamma+\xi\beta)}$. Comparing $\underline{q}^V_{2t}$ with $\overline{q}^V_t$ and rearranging give $\underline{q}^H < \underline{q}^H_{1t} = \frac{\mu[2k\lambda^2(\gamma+\xi\beta)-(\beta+\gamma)]}{k[\mu-\lambda^2(\beta+\gamma)(\gamma+\xi\beta)]}$. Therefore, when $\underline{q}^H < \underline{q}^H_{1t}$, the threshold $\underline{q}^V_{2t}$ lies within the feasible region. It follows that $q^{V*} > q^{H*}$ when $\underline{q}^V > \underline{q}^V_{2t}$, whereas $q^{V*} < q^{H*}$ when $\underline{q}^V < \underline{q}^V_{2t}$. By contrast, when $\underline{q}^H > \underline{q}^H_{1t}$, we have $\underline{q}^V_{2t} \geq \overline{q}^V_t$. Because every feasible $\underline{q}^V$ satisfies $\underline{q}^V < \overline{q}^V_t$, there exists no feasible $\underline{q}^V$ satisfying $\underline{q}^V > \underline{q}^V_{2t}$. Consequently, $q^{V*} < q^{H*}$ throughout the feasible region.

Part (c).

From Lemma A6, we have $R^{H*} = \frac{\mu[\beta+\gamma+k\underline{q}^H]^2}{2[2k\mu-(\beta+\gamma)^2]}$ and $R^{V*} = \frac{\mu[\underline{q}^V+\lambda^2(\beta+\gamma)]^2}{2\lambda^2[2\mu-\lambda^2(\beta+\gamma)^2]}$. Under the interior-equilibrium conditions, all terms inside the squared expressions are positive. Therefore, $R^{V*} > R^{H*}$ is equivalent to $\frac{\underline{q}^V+\lambda^2(\beta+\gamma)}{\lambda\sqrt{2\mu-\lambda^2(\beta+\gamma)^2}} > \frac{\beta+\gamma+k\underline{q}^H}{\sqrt{2k\mu-(\beta+\gamma)^2}}$. Solving this inequality for $\underline{q}^V$ gives $\underline{q}^V > \underline{q}^V_{3t} = \lambda\sqrt{\frac{2\mu-\lambda^2(\beta+\gamma)^2}{2k\mu-(\beta+\gamma)^2}}[\beta+\gamma+k\underline{q}^H]-\lambda^2(\beta+\gamma)$. Hence, the platform-profit comparison is characterized by a unique threshold in $\underline{q}^V$. Comparing $\underline{q}^V_{3t}$ with the feasible upper bound $\overline{q}^V_t$ and rearranging give $\underline{q}^H < \underline{q}^H_{2t} = \frac{1}{k}\{\frac{\lambda(\gamma+\xi\beta)\sqrt{[2\mu-\lambda^2(\beta+\gamma)^2][2k\mu-(\beta+\gamma)^2]}}{\mu-\lambda^2(\beta+\gamma)(\gamma+\xi\beta)}-(\beta+\gamma)\}$. When $\underline{q}^H < \underline{q}^H_{2t}$, the threshold $\underline{q}^V_{3t}$ lies within the feasible region. Therefore, $R^{V*} > R^{H*}$ when $\underline{q}^V > \underline{q}^V_{3t}$, whereas $R^{V*} < R^{H*}$ when $\underline{q}^V < \underline{q}^V_{3t}$. By contrast, when $\underline{q}^H > \underline{q}^H_{2t}$, we have $\underline{q}^V_{3t} \geq \overline{q}^V_t$. Thus, every feasible $\underline{q}^V$ satisfies $\underline{q}^V < \overline{q}^V_t \leq \underline{q}^V_{3t}$, which implies $R^{V*} < R^{H*}$ throughout the feasible region.

Part (d).

For the virtual avatar streamer, differentiating S^{V*} with respect to $\underline{q}^V$ gives $\frac{\partial S^{V*}}{\partial \underline{q}^V} = \frac{\mu^2(\beta+\gamma)}{[2\mu-\lambda^2(\beta+\gamma)^2]^2} + \frac{[\lambda^2(\beta+\gamma)^2-\mu][3\mu-\lambda^2(\beta+\gamma)^2]}{\lambda^2[2\mu-\lambda^2(\beta+\gamma)^2]^2}\underline{q}^V - 1$. It is easy to find that $\frac{\partial S^{V*}}{\partial \underline{q}^V} < 0$ for every feasible $\underline{q}^V > 0$. Hence, S^{V*} is continuous and strictly decreasing in $\underline{q}^V$. For the human streamer, differentiating S^{H*} with respect to $\underline{q}^H$ gives $\frac{\partial S^{H*}}{\partial \underline{q}^H} = \frac{k\{k\mu^2(\beta+\gamma)-[k\mu-(\beta+\gamma)^2][3k\mu-(\beta+\gamma)^2]\underline{q}^H\}}{[2k\mu-(\beta+\gamma)^2]^2}$, and $\frac{\partial^2 S^{H*}}{\partial(\underline{q}^H)^2} = -\frac{k[k\mu-(\beta+\gamma)^2][3k\mu-(\beta+\gamma)^2]}{[2k\mu-(\beta+\gamma)^2]^2} <$

0. Thus, S^{H*} is strictly concave in $\underline{q}^H$ and initially increases with $\underline{q}^H$. Then, we have $S^{H*}(\underline{q}^H = 0) = \frac{k\mu^2(\beta+\gamma)^2}{2[2k\mu-(\beta+\gamma)^2]^2}$ and $S^{V*}(\underline{q}^V = 0) = \frac{\lambda^2\mu^2(\beta+\gamma)^2}{2[2\mu-\lambda^2(\beta+\gamma)^2]^2}$. It is easy to find that $S^{H*}(\underline{q}^H = 0) > S^{V*}(\underline{q}^V = 0)$ under $k\lambda^2 < 1$. Together with $S^{H*}(\overline{q}_t^H) < S^{V*}(\underline{q}^V = 0)$ and the strict concavity of S^{H*}, this implies that there exists a unique threshold $\underline{q}_{3t}^H \in (0, \overline{q}_t^H)$ satisfying $S^{H*}(\underline{q}_{3t}^H) = S^{V*}(\underline{q}^V = 0)$. When $\underline{q}^H < \underline{q}_{3t}^H$, we have $S^{H*} > S^{V*}(\underline{q}^V = 0)$. Since S^{V*} is strictly decreasing in $\underline{q}^V$, it follows that $S^{H*} > S^{V*}$ for every feasible $\underline{q}^V > 0$. When $\underline{q}^H > \underline{q}_{3t}^H$, we have $S^{V*}(\underline{q}^V = 0) > S^{H*} > 0$. Because S^{V*} is continuous and strictly decreasing in $\underline{q}^V$ and satisfies $S^{V*}(\overline{q}_t^V) < 0 < S^{H*}$, there exists a unique threshold $\underline{q}_{4t}^V \in (0, \overline{q}_t^V)$ satisfying $S^{V*}(\underline{q}_{4t}^V) = S^{H*}$. Therefore, when $\underline{q}^H > \underline{q}_{3t}^H$, we have $S^{V*} > S^{H*}$ if $\underline{q}^V < \underline{q}_{4t}^V$, whereas $S^{V*} < S^{H*}$ if $\underline{q}^V > \underline{q}_{4t}^V$. □